

[2] Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

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1-Minute Executive Takeaway

Key Finding: The seminal paper establishing that external retrieval dramatically reduces hallucination in generative language models.

Core Research Objective

To develop a general-purpose hybrid architecture that combines parametric memory (pre-trained seq2seq models) with non-parametric memory (dense vector index of Wikipedia) for knowledge-intensive generation tasks.

Methodology & Architecture

Proposed RAG-Sequence and RAG-Token models using a dense passage retriever (DPR) coupled with a BART generator, fine-tuning retrieval and generation end-to-end.

Key Empirical Findings & Evidence

RAG models achieved state-of-the-art results on open-domain question answering (Natural Questions, TriviaQA), substantially reducing factual hallucinations compared to purely parametric models.

Direct Relevance to Our Kwiz Paper

Foundational citation in Section 1 and Section 2.1. Our architecture adapts Lewis et al.'s retrieval-augmented paradigm specifically for educational courseware, using local dense retrieval (`nomi-embed-text`) to ground programming questions in instructor-uploaded slides.